Implementation Roadmap

Hasnahena Platform

Version:	1.0
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Status:	Pre-Coding Phase
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Execution plan from project setup
to production-ready MVP

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Contents

Document Control

Item	Details
Document Title	Implementation Roadmap
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Revision History

Ver.	Date	Author	Changes
1.0	March 22, 2026	Md Saif Ahmed	Initial release

1 Objective

This document provides a pragmatic execution roadmap from project setup to production-ready MVP based on finalized requirements. The roadmap is designed to:

- Deliver working software in incremental, testable chunks
- Prioritize core financial flows that provide immediate value
- Minimize risk through early validation of critical paths
- Enable parallel workstreams where dependencies allow

2 Workstream Breakdown

Workstream	Deliverables
Foundation	Repository setup, coding standards, CI/CD skeleton, environment templates, development tooling
Identity & RBAC	Authentication module, role middleware, session/token security policies, password management
Member Lifecycle	Registration form, approval workflow, first login flow, nominee management, profile updates
Core Finance	Deposit/withdrawal/loan/donation APIs, transaction state machine, ledger integrity
Operations	Worker queue tools, admin controls, dispute and correction mechanisms, SLA tracking
Observability	Audit log implementation, basic metrics, operational alerts, backup procedures

3 Increment Plan

3.1 Increment 1: Foundation & Identity

Focus: Project infrastructure and authentication layer

Deliverables:

- Repository structure with monorepo or multi-repo setup
- CI/CD pipelines for frontend and backend
- Database schema baseline with migrations
- Authentication (login, logout, password change)
- RBAC middleware for Member, Worker, Admin roles
- Basic landing page with registration form UI

Exit Criteria:

- Users can register, login, and access role-appropriate pages

- RBAC prevents unauthorized endpoint access
- CI/CD deploys to staging on merge

3.2 Increment 2: Deposit & Withdrawal

Focus: Core financial flow for savings

Deliverables:

- Deposit request API and UI
- Worker processing interface
- Member confirmation flow
- Withdrawal request with notice period validation
- Transaction ledger with proper accounting
- Member dashboard showing balances

Exit Criteria:

- Complete Request → Process → Confirm cycle works
- Ledger balances are accurate
- Withdrawal rules enforced

3.3 Increment 3: Loan Lifecycle

Focus: Loan application, approval, and repayment

Deliverables:

- Loan eligibility calculation (3-cap system)
- Loan application form and API
- Guarantor request and approval workflow
- Admin loan approval/rejection
- Worker loan disbursal and member confirmation
- Installment recording

Exit Criteria:

- Member can apply for loan within eligibility
- Guarantor workflow functions correctly
- Loan states transition correctly through lifecycle

3.4 Increment 4: Donations & Disputes

Focus: Donation management and error handling

Deliverables:

- Donation form with fund type selection
- Donation fund tracking (Operating, Lending, Zakat, Sadaqah)
- Dispute raising mechanism
- Admin dispute resolution workflow
- Default settlement waterfall implementation

Exit Criteria:

- Donations correctly categorized and tracked
- Disputes freeze transactions and alert admin
- Settlement waterfall executes correctly

3.5 Increment 5: Polish & Release

Focus: Dashboards, hardening, and release preparation

Deliverables:

- Member dashboard with all summaries
- Worker dashboard with pending queues and SLA indicators
- Admin dashboard with approvals, alerts, and settings
- Audit log viewer for admin
- Localization (English/Bengali) completion
- Security hardening and penetration testing
- Documentation and training materials

Exit Criteria:

- All dashboards functional and accurate
- No critical/high security vulnerabilities
- System ready for production deployment

4 Readiness Checklist

Before development begins, the following must be complete:

4.1 Documentation Readiness

Business Requirements Document approved

Software Requirements Specification approved

Functional Requirements Document approved

API Contract Specification approved

Database Schema Design approved

Test Plan agreed by engineering and operations

4.2 Infrastructure Readiness

Source control repositories created

CI/CD pipeline templates ready

Development environment provisioned

Staging environment provisioned

Database instances created (dev/staging)

Environment variables documented

4.3 Team Readiness

Development team assigned

Roles and responsibilities defined

Communication channels established

Sprint cadence agreed

5 Risk Mitigation

Risk	Impact	Mitigation
Financial calculation errors	Critical	Extensive unit tests, double-entry validation
RBAC bypass	Critical	Security testing, code review gates
Data loss	Critical	Automated backups, transaction logging
Scope creep	High	Strict increment boundaries, change control
Integration delays	Medium	Mock services, contract testing

6 Approvals

Product Owner	**Technical Lead**
Date: _____	Date: _____